

Bilal Ahmad Kalas

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Profile

I am an M.Sc. Mathematics graduate with research interests in algebra and its interactions with topology, geometry and number theory. My academic work has focused on finite transformation semigroups and independent study in pure mathematics. I am seeking doctoral opportunities in pure mathematics with particular interests in algebra, commutative algebra, algebraic geometry, arithmetic geometry, algebraic topology and related areas.

Education

Integrated B.Sc.-M.Sc. Mathematics
Central University of Kashmir, India

2019–2024

Master's Dissertation

Dissertation: Ranks and Presentation for Finite Transformation Semigroups

[Link](#)

Supervisor: Dr Aftab Hussain Shah (Associate Professor)

- Studied rank and presentation problems for finite order-preserving transformation semigroups.
- Based on the work of Joerg Koppitz and Somuek Worawiset on transformations with one fixed point.
- Focused on semigroup presentations, generating set, ranks, and structural & combinatorial properties of finite transformation semigroups.

Research Interests

Algebra, Commutative Algebra, Galois Theory, Representation Theory, Algebraic Topology, Algebraic Number Theory, Algebraic Geometry, Semigroup Theory, Topology

Teaching Experience

Private Mathematics Tutor

2024-2025

Skills

- Relevant Coursework: Abstract Algebra, Commutative Algebra, Galois and Field Theory, Topology, Number Theory, Semigroup Theory
- Academic Skills: Mathematical proof writing, mathematical exposition, independent study, reading and analysis of mathematical literature.
- Technical skills: Latex, Python, HTML/CSS, JavaScript, C

Languages

- Hindi/Urdu – Native
- Gojri – Native
- Kashmiri – Native
- English – Fluent

References
